



Fix the Turtle Code

Ontario Math C3.2 — find and fix code

Name: _____

Date: _____

★ I can find the bug in turtle code and fix it.

Find it, then fix it

Sometimes turtle code draws the wrong thing. A coder reads it, decides what it SHOULD draw, and finds the wrong number.

A turtle-loop bug is usually the wrong range number (too few/many sides) or the wrong turn. Help Bit fix them.

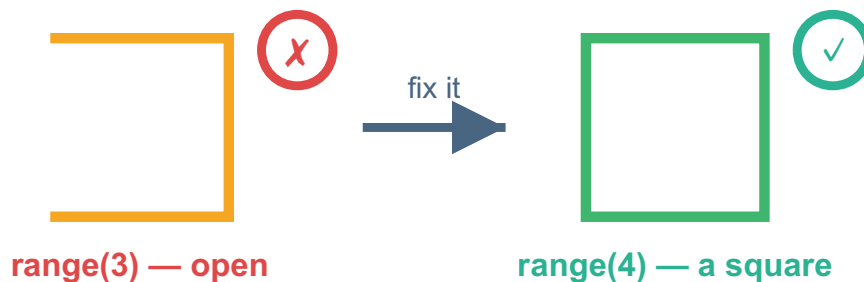
Bit should draw a SQUARE — but this is buggy

PYTHON

```
for i in range(3):  
    bit.forward(50)  
    bit.right(90)
```

OUTPUT

Only 3 sides — the square does not close.



range(3) leaves the square open — fix it to range(4).

1 Find the bug

Bit should draw a square, but the code says range(3). What range number fixes it? range(_____)

2 Fix the turn

Bit should draw a TRIANGLE, but the code is: `for i in range(3): bit.forward(50); bit.right(90)`. The turn is wrong. What should `right()` be? (Hint: a triangle turns 120.)

3 Fix the count

Bit should draw a straight line 100 long, moving 20 each time. The code says `range(3)`. What range number fixes it? `range(_____)`

4 Challenge

Bit draws a square with `range(4)`, `forward(50)`. To make a BIGGER square, a student changed `range(4)` to `range(8)`. That did NOT work. What should they change instead, and why?

Big idea

To debug turtle code: say what it SHOULD draw, then check the range (how many sides) and the turn. Change the one that is wrong — to resize, change forward, not the range.